

Flying Liferaft

Detailed Modelling and Control of the VTOL and Transition Flight Segments



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Abstract

This report aims to validate the use of 4 vertically aligned propellers to achieve vertical takeoff and landing and transition from hover to forward flight. Given the current design parameters, this was found to be achievable and optimised control gains, as well as a detailed method to obtain these, are presented. PID control was utilised due to its relative ability to track both step and ramp signals, both of which are required. However, the results also suggest that the aircraft is currently difficult to control given the current estimates for motor spool up time and longitudinal CG location, suggesting that improvement of both these parameters is required in further design stages.

Contents

List of Figures	v
List of Tables	vi
1 Introduction	1
2 Equations of motion	1
2.1 Fixed-body axes	1
2.2 Rigid body assumption	2
2.3 VTOL propellers	2
2.4 Transition	2
3 Flight path definition	2
4 Modelling	3
4.1 Nonlinear vs linearised model	3
4.2 Motor modelling	3
4.3 Drag in VTOL	4
5 Controller	4
5.1 Controller choice	4
5.2 Altitude controller	4
5.3 Pitch, roll and yaw controllers	5
5.4 Overall controller	6
6 Optimisation of control gains	6
6.1 Optimisation method	6
6.1.1 Aggression parameter	6
6.2 Altitude control gains optimisation	6
6.3 Pitch control gains optimisation	7
6.4 Roll and yaw control gains	9
6.5 Summary of control gains	10
7 Results	10
7.1 Transition	10
7.2 Takeoff and landing	11
8 Conclusion	12
8.1 CG location	12
8.2 Motor spool-up time	12
References	12

List of Figures

1	Controller performance for the optimised control gains. W refers to the climb rate here due to the negligible pitch or roll angles.	7
2	Sorted cost function from high to low for the last 300 simulations for the altitude controller. The optimal point is the most right point on the plot.	7
3	Pitch controller performance	8
4	Sorted cost function utilising equation (19)	8
5	Pitch controller performance	9
6	Sorted cost function utilising equation (20)	9
7	Roll controller performance.	10
8	Response of the aircraft to a simultaneous roll and pitch perturbation. The colours for the throttle lines use the same layout as the legend dictates in figure 7.	10
9	Performance of the aircraft in the forward transition (from hover to forward flight). .	11
10	Performance of the aircraft in the reverse transition (from forward flight to hover). .	11
11	Performance of the aircraft in takeoff and landing.	12

List of Tables

1	Fixed-body variable names and symbols.	1
2	Inertial values for the aircraft [1].	1
3	Optimised control gains for the altitude controller relative to the cost function given by equation (18).	7
4	Optimised control gains for the pitch controller relative to the cost functions $Q_{\Theta,1}$ and $Q_{\Theta,2}$ in equations (19) and (20) respectively.	9
5	Roll control gains and associated performance values as seen in figure 7.	10
6	Optimised control gains for the pitch controller relative to the cost functions $Q_{\Theta,1}$ and $Q_{\Theta,2}$ in equations (19) and (20) respectively.	10

1 Introduction

The use of a quadcopter style layout to achieve vertical takeoff and landing (VTOL) presents a complex and challenging modelling and control problem. The reason for this is that the dynamics of such a system are nonlinear, requiring a more complex model and control system. This report aims to analyse, with a reasonable degree of detail, the ability of this aircraft to achieve takeoff, landing and transition between hover and forward flight, as well as presenting a detailed method for optimising the control gains used.

2 Equations of motion

2.1 Fixed-body axes

The choice was made to utilise a fixed-body axis system for this problem, with the symbols used in this system expressed in table 1. The key feature of this axis system is that it lies in the frame of reference of the aircraft rather than remaining in an earth-centric frame of reference. There were two key reasons for this choice of axes. Firstly, the equations of motion for a fixed-body axis system are well established and are readily available. The second reason was that, by aligning the axes with the zero angle of attack of the aircraft, the VTOL propeller thrust can be considered to act only in the Z direction. This simplifies the analysis significantly in further steps and also allows for the approximation that, at steady state and with zero side-slip, the pitch angle is equal to the angle of attack of the aircraft. As an additional benefit, the pitch, roll and yaw moments generated by the VTOL propellers can be considered to be constant in this frame of reference provided the next key assumption holds. The equations of motion in this axis system are given in equation (1), where the majority of symbols have the same meaning as in table 1 [2]. The navigation equations listed in [2] were also implemented in order to measure the earth-relative altitude of the aircraft. The variable m represents the aircraft mass and I_{xx} , I_{yy} , I_{zz} and I_{xz} the moments of inertia. The values for these variables can be found in table 2 [1].

Table 1: Fixed-body variable names and symbols.

Component	Velocity	Euler angle and derivatives	Angle name	Forces	Moments
i	U	Φ, p	Roll	X	L
j	V	Θ, q	Pitch	Y	M
k	W	Ψ, r	Yaw	Z	N

$$\begin{aligned}
 X &= m(\dot{U} - Vr + Wq) \\
 Y &= m(\dot{V} + Ur - Wp) \\
 Z &= m(\dot{W} - Uq + Vp) \\
 L &= I_{xx}\dot{p} - I_{xz}\dot{r} + (I_{zz} - I_{yy})qr - I_{xz}pq \\
 M &= I_{yy}\dot{q} + (I_{xx} - I_{zz})pr + I_{xz}(p^2 - r^2) \\
 N &= I_{zz}\dot{r} - I_{xz}\dot{p} + (I_{yy} - I_{xx})pq + I_{xz}qr
 \end{aligned} \tag{1}$$

Table 2: Inertial values for the aircraft [1].

m	51.5 kg
I_{xx}	6.8024 kg.m ²
I_{yy}	5.0295 kg.m ²
I_{zz}	10.4437 kg.m ²
I_{xz}	-0.1555 kg.m ²

2.2 Rigid body assumption

The use of the rigid body assumption results in a few key simplifications and requirements for the controller. Firstly, the pylons that the VTOL propellers are connected to can be assumed to remain fixed for the duration of the mission profile, meaning that the thrust from each propeller can always be assumed to act in the same direction. Through an FEA analysis (REFERENCE), it was found that the maximum deflection and slope of the pylon at the position of the propellers was VALUE and VALUE. This results in a decrease in thrust acting in the Z direction of approximately 0.01%, meaning it can reasonably be neglected under normal operating conditions. This assumption also implies that any aeroelastic effects can be ignored.

However, the importance of normal operating conditions cannot be neglected. Specifically, oscillatory motion of the aircraft could cause this assumption to become invalid especially if the frequency of oscillations was near the aircraft's natural frequency. As the aircraft's response to oscillatory loads is not available at this stage of the design, the controller design will attempt to avoid high frequency oscillations when correcting perturbations.

2.3 VTOL propellers

The fixed-body axes system and rigid body assumption provides a convenient simplification to the forces imparted on the aircraft by the 4 VTOL propellers. Namely, the X and Y components will always be zero as the thrust will always act in the negative Z direction. This then means that the fixed-body forces can be expressed as shown in equation (2), where T_i represents the thrust generated by the i^{th} propeller. The rolling, pitching and yawing moments generated by the propellers can also be expressed as shown in equation (3), where x_i is the position of the i^{th} propeller relative to the nose of the aircraft, x_{CG} is the longitudinal CG location relative to the nose of the aircraft, y_i is the lateral position of the i^{th} propeller relative to the centre line of the aircraft and P_i and ω_i are the power and rotational velocity generated by the i^{th} propeller. A plus or minus symbol is used in the yawing equation as this value depends on the direction of rotation of the propellers. As the aircraft is designed to be symmetric about the centre plane, these rotation directions are such that, in both pitch and roll, the yawing moment will still be zero. From here onwards, looking at the aircraft from the nose, propeller 1 refers to the front left propeller, propeller 2 refers to the front right propeller, propeller 3 refers to the back left propeller and propeller 4 refers to the back right propeller. The positions can be found in the layout analysis of the aircraft [1].

$$X = 0, Y = 0, Z = -\sum T_i \quad (2)$$

$$\begin{aligned} L &= \sum T_i y_i \\ M &= \sum T_i (x_i - x_{CG}) \\ N &= \sum \pm P_i / \omega_i \end{aligned} \quad (3)$$

2.4 Transition

The transition is defined as the flight segment where the aim is to move from a steady hover to forward flight. This is achieved by increasing forwards velocity from zero to cruise speed, causing the generation of upwards force to transition from the VTOL propellers to the lifting surfaces. In order to model the point at which this occurs, the dynamic stability derivatives are used [3] to model the aerodynamic forces generated by the aircraft in forward flight. Fortunately, these derivatives are evaluated about an angle of attack of zero. However, they are evaluated at cruise velocity. As only the symmetric stability derivatives are used, this issue can be overcome by simply scaling the derivatives by the current U velocity squared divide by the cruise speed of 58.3 m/s squared. Performing this scaling should provide a valid estimation of the aerodynamic forces and moments produced as the aircraft enters forward flight, though caution must be taken if the response of the aircraft to perturbations mid-transition is ever to be tested.

3 Flight path definition

The aim of this report is to analyse specifically the VTOL and transition performance. As such, a flight path defining both segments must be created. These flight paths are denoted as r as they

function as a reference signal in the model. This vector contains 5 entries in the form shown in equation (4), where h is the altitude in metres, Θ , Φ and Ψ are the pitch, roll and yaw angles respectively measured in radians and V_∞ is the free-stream velocity measured in m/s. For the takeoff, a step altitude input of 12 m is to be used, with all other entries being set to zero. In landing, a descent rate of 2 m/s was chosen. As a step input would result in the controller turning off the propellers completely, a ramp with gradient 2 m/s is used here, again with all other states being set to zero. The transition uses the same 12 m altitude, but sets the pitch angle reference to 0.1 degrees and the freestream velocity to the cruise speed of 58.3 m/s. This will cause the pusher propeller to turn on and the VTOL propellers will achieve the angle of 0.1 degrees, allowing sufficient lift to be generated in order to transition [?]. The transition from forward flight to hover is slightly more complicated, as the controller in this report will not control for velocity but instead for pitch angle. As such, in order to slow down from 10 m/s to stationary whilst maintaining a 12 m altitude, a pitch angle reference of 7.2 degrees is applied for 10 seconds. This sufficiently reduces the velocity from 10 m/s to zero as will be seen later.

$$r = [h \quad \Theta \quad \Phi \quad \Psi \quad V_\infty]^T \quad (4)$$

4 Modelling

4.1 Nonlinear vs linearised model

In order to linearise the aircraft, a stable equilibrium point is required so that a Jacobian for the system can be evaluated. In this case, the definition of stable equilibrium is a point about which the aircraft will return to if a relatively small perturbation is applied. The two equilibrium conditions for this aircraft are a constant climb and hover, specifically because the mission profile does not ask for any forwards or sideways translation by use of the VTOL propellers. In the case of a small forwards or sideways velocity perturbation, the aircraft will return to the equilibrium velocity of zero due to the drag of the aircraft, neglecting other aerodynamic forces. However, in a pitch, roll or yaw perturbation, the aircraft will not naturally return to the equilibrium as the pitching moment generated by the propellers will remain at zero within the aircraft's frame of reference. As such, linearising the model in this case is not possible and the choice was instead made to utilise the nonlinear form of the equations in equation (1).

4.2 Motor modelling

The aim of the controller will be to prescribe a target RPM, Ω_T , to each VTOL propeller in order to correct perturbations. It is unlikely that the motor will be able to instantaneously reach this target RPM and will instead steadily approach the target value. Whilst a typical quadcopter can be expected to have motor spool up times which are almost instant due to their lightweight design, this aircraft is much heavier and therefore requires much larger blades. Whilst achieving similarly fast spool up times would be achievable through the use of a very powerful motor, this would come with the penalty of requiring a more powerful battery, increasing weight. This increase in weight would then result in a requirement for a larger propeller, higher maximum RPM or both, meaning that the expected increase in spool up time may be less than expected.

In modelling the motor, some reasonable assumptions were used. Firstly, the rate at which the RPM increases was expected to lower as the propeller approached the RPM, yielding the relation shown in equation (5). Secondly, the time taken to reach 99% of the maximum target RPM should be 5 seconds. This value was chosen based on initial estimates for the propeller size and using some degree of intuition. For this reason, a rough robustness analysis will need to be conducted at a later stage to suggest a maximum allowable spool up time for the controller gains which will be evaluated in order to maintain controllability. Through the analysis shown in equation (6) and using the 5 second choice, the value for T was found to be approximately 1.0857. By taking the Laplace transform of equation (5), the transfer function for RPM ($\Omega(s)$) and throttle ($T(s)$) can be seen in equation (7), where 6000 is used in the numerator as this is the maximum achievable RPM for the motor.

$$\Omega(t) = \Omega_T \left(1 - \exp\left(-\frac{t}{T}\right) \right) \quad (5)$$

$$1 - \exp\left(-\frac{5}{T}\right) = 0.99 \implies T = \frac{\ln(100)}{5} \approx 1.0857 \quad (6)$$

$$\frac{\Omega(s)}{T(s)} = \frac{6000}{Ts + 1} \approx \frac{1}{1.0857s + 1} \quad (7)$$

As an extension of the motor modelling, it must also be taken into account that the thrust, power and, as a result, yawing moment generated by each propeller is a function of both the RPM and the current climb velocity. Utilising data obtained through analysis of the propellers [1], 2D lookup tables were implemented in Simulink in order to model this behaviour.

4.3 Drag in VTOL

An important factor is the drag generated by the aircraft when climbing vertically. This figure is quite difficult to obtain without real world testing as there are many complex factors which will add to the drag vertically. Without going into detail, the way of modelling this drag was done by obtaining climb rate data from the X-Plane flight simulation when a 100% step throttle is applied to the VTOL propellers and the CG is placed at the centre of the propellers so as to avoid any pitching behaviour. A balance of forces was then conducted over the period where the acceleration of climb rate was relatively constant. As the value for the drag coefficient is essentially linked to the reference area chosen for this calculation, the decision was instead taken to evaluate the product of drag coefficient and area such that this parameter could simply be multiplied by the vertical dynamic pressure in order to obtain the drag. This value was found to be 1.6707 m^2 and this was then implemented into the model.

5 Controller

5.1 Controller choice

Literature review suggested that a typical choice for a drone controller is a proportional-derivative (PD) controller. However, in this case, such a controller would not be suitable as the steady state error would not be acceptable when attempting to track a ramp signal. Similarly, due to the small altitude to which the drone is climbing, the steady state error when tracking a step reference of may not be acceptable due to obstacles such as ships and waves. If the decision were also made to deliver the payload during a VTOL hover, a steady state tracking error could also mean the difference between dropping the payload from a very high altitude or injuring the people stranded at sea. As such, a proportional-integral-derivative controller (PID) was chosen. This control law can be seen in equation 8, where u is the control term, e is the error signal and K_p , K_i and K_d are proportional, integral and derivative control gains respectively. For sake of completeness, the Laplace transform or frequency domain version of the PID control equation is given by equation (9). An important distinction to note here is the use of the filter coefficient N in the derivative term. This variable acts as a low pass filter for the derivative term, allowing derivatives far below N to pass and attenuating derivatives close to or larger than this value.

$$u = K_p e + K_i \int e dt + K_d \frac{de}{dt} \quad (8)$$

$$U(s) = E(s) \left(K_p + K_i \frac{1}{s} + K_d \frac{Ns}{N + s} \right) \quad (9)$$

5.2 Altitude controller

For the altitude controller, the error signal e_h refers to the difference between the reference altitude r_h and the actual altitude y_h and is given by equation (10). The altitude y_h is simply given by the negative earth-relative Z distance from the navigation equations seen discussed prior and the reference altitude was set to be 12 m based on obstacle and wave height considerations [1]. Under the assumption that the CG lines up with the centre of the 4 VTOL propellers, the thrust generated by each propeller must be equal in order to maintain the same orientation and enter a pure climb. As such, the relation between control signal u and target RPM ω is given by equation (11). In

reality, the CG will not line up with the centre of the 4 propellers. However, assuming the pitch and roll controllers are designed well enough to counter this, this relation for altitude control is anticipated to still work. The altitude control term produced was saturated between 0 and 1. This decision is based on the fact that the most the controller can do is either prescribe full throttle to achieve an altitude or turn off the propellers if the aircraft is sufficiently above altitude.

$$e_h = r_h - y_h \quad (10)$$

$$\omega = \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}^T u_h \quad (11)$$

5.3 Pitch, roll and yaw controllers

The error signal for pitch, e_θ , can be given by equation (12), where r_θ and y_θ refer to the reference and actual pitch angle respectively. Rather than assuming that the longitudinal CG, x_{CG} , lies at the centre of the front and aft propellers, the control law will be generalised for an arbitrary CG location. The reason for this is that, as the CG moves, the pitching moment generated by the front and aft propellers will vary and the relative change in thrust required for each propeller will be different. This can simply be shown by the a pitching moment calculation given in the equation for pitching moment M in equation (3). As the relative change in thrust is related to the distance between the propellers are the CG, the relation between pitch change in RPM and pitch control term is given by equation (13), where x_F and x_A refer to the longitudinal position of front propellers (1 and 2) and aft propellers (3 and 4) respectively. The choice of multiplying by 2 is arbitrary, but is simply chosen by noting the fact that, if the CG lies at the centre of the two propellers, the vector will be fully normalised as all entries will be ± 1 . The pitch control input was saturated between -0.5 and 0.5. The reason for this was that, in a scenario where a large pitch perturbation is seen, the controller will prescribe a throttle to minimise one of the front or aft propeller pair and maximise the other. 0.5 was chosen as a reasonable value to allow the pitch controller to accomplish this result without excessively reducing the throttle on one side and increasing the throttle on the other, as this could amplify perturbations in the system if the controller becomes too aggressive.

$$e_\theta = r_\theta - y_\theta \quad (12)$$

$$\omega = \frac{2u_\theta}{x_F - x_A} \begin{bmatrix} x_F - x_{CG} & x_F - x_{CG} & x_A - x_{CG} & x_A - x_{CG} \end{bmatrix}^T \quad (13)$$

Following a similar line of reasoning, the roll controller can be by the error signal given in equation (14). As the aircraft is assumed to be designed such that it is symmetric about the center plane, the equation relating the control term and the prescribed RPM is given by equation (15). For the same reasons as stated above, the roll controller was saturated at ± 0.5 . Finally, the yaw controller error term and RPM can be described by equations (16) and (17). This equation accounts for the rotation direction of the 4 propellers as chosen by the propulsion team [1]. The yaw controller was saturated between ± 0.1 as the yaw controllability is much less important than the roll and pitch controllers, and limiting the control input it can impart was presumed to be productive in the context of maximising roll and pitch performance. Though not discussed, this was found to be correct, as a roll or pitch perturbation reduces the thrust acting upwards possibly resulting in the aircraft crashing. Ensuring that the controller prioritises roll and pitch perturbations over yaw perturbations therefore significantly reduces the risk of a huge loss in altitude due to perturbations in the aircraft's orientation.

$$e_\phi = r_\phi - y_\phi \quad (14)$$

$$\omega = \begin{bmatrix} 1 & -1 & 1 & -1 \end{bmatrix}^T u_\phi \quad (15)$$

$$e_\psi = r_\psi - y_\psi \quad (16)$$

$$\omega = \begin{bmatrix} 1 & -1 & 1 & -1 \end{bmatrix}^T u_\psi \quad (17)$$

5.4 Overall controller

The overall VTOL controller involves summing up the contributions from all of the altitude, pitch, roll and yaw controllers. The resulting summation is then saturated a final time between 0 and 1. The result of all the control inputs evaluated above is a throttle which will aim to counter all perturbations simultaneously, which prompts a concern about the performance during initial takeoff. If the aircraft suffers a simultaneous pitch and roll oscillation near the ground, the controller, in trying to correct both these perturbations and climb to the reference altitude, could end up crashing. It may therefore be beneficial to implement a fail safe near the ground where maximum throttle is applied if a minimum altitude is reached to attempt to ascend high enough before attempting to correct the perturbations. Though this was not studied, further investigations of performance may require such a fail safe.

6 Optimisation of control gains

The use of the VTOL propellers requires a significant amount of power which in turn requires a larger battery, incurring a significant weight penalty. Given that the total energy consumed is also a function of time, the amount of time taken to reach a target altitude and successfully transition from VTOL to forward flight is also important. With these considerations in mind, it is important to highlight the need for some form of controller optimisation such that the power consumed is minimised, leaving a reasonable amount of excess power for pitch, roll and yaw controllability and ensuring that the chosen battery is sufficient.

6.1 Optimisation method

Each of the 4 states which are to be controlled by the VTOL propellers utilise PID control. As such, they each have 3 control gains associated with them. As this results in 12 gains which need to be tuned and a filter coefficient for each PID, a more efficient process for tuning each of them is required. The chosen approach was to tune the altitude control gains using a pure takeoff to the required altitude, and tune each of the rotation control gains using a angular rate perturbation. Thousands of possible combinations of the control gains are then spawned and a cost function, referred to hereafter as Q , is then optimised. These results are filtered with respect to a specified cutoff region and new control gain ranges are implemented. This process is repeated until an optimum set of control gains with respect to that particular cost function is obtained.

6.1.1 Aggression parameter

The throttle command prescribed by the controller has the potential to be very oscillatory. Such a controller is referred to hereafter as aggressive. This behaviour can occur for many reasons, but is largely a combination of noise in the signal being amplified by the derivative term and the spool up time of the motor, causing the throttle to act as a pulse width modulator attempting to maintain relatively constant thrust by way of a high frequency oscillatory throttle. This behaviour is not only unrealistic as the aim would be to maintain constant throttle input for a hover, but an oscillatory throttle can result in oscillatory behaviour in states. For example, an altitude controller which prescribes an aggressive throttle could result in an oscillatory climb velocity. If these oscillations in climb rate were large or near the natural frequency of the aircraft, the rigid body assumption would no longer hold and effects such as aeroelasticity could no longer be ignored. An aggressive controller can be improved simply by changing (usually increasing) the value of the filter coefficient, meaning that the optimisation involves an extra step of altering the filter coefficient for each new set of optimal control gains before running a new batch of simulations.

6.2 Altitude control gains optimisation

In order to purely optimise the climb performance, the longitudinal CG is placed at the centre of the two propellers in order to prevent pitching. The actual CG location will be implemented when tuning the pitch controller and the full range of CG locations will be tested in a robustness analysis, but at this stage such a consideration is not necessary and would complicate the tuning process for the altitude controller.

Based on performance calculations [1], the anticipated energy required and time taken to perform a takeoff are 120 kJ and 15 seconds respectively. Based on these parameters, the cost function Q is proposed in equation (18), where T_s is the time taken from leaving the ground to achieve a stable altitude and E is the total energy used over this period. The time and energy are calculated relative to leaving the ground as a controller which takes a long time to takeoff due to the motor spool up time may actually display better settling behaviour, which is advantageous for tracking a more complex reference signal. The aircraft is defined as reaching a stable altitude once the altitude error drops to below 10 cm and remains there for the remaining duration of the simulation. Simulations where this behaviour is not observed are classified as unstable and are removed from the optimisation. The cutoff region for this optimisation was set to be at a cost value of 2, as the controller must not take longer and not consume more energy than the values stated above, meaning both terms in the cost equation will be less than 1.

After obtaining rough estimates for the PID control gains, many simulations were tested to try and obtain control gains which optimised this cost function. After each batch of simulations were executed, the optimum set of gains based on the cost function below was extracted and the filter coefficient was varied until oscillations in the throttle command and climb velocity were minimised. This process was repeated until the gains and associated cost parameters presented in table 3 were obtained. The resulting climb performance and throttle can be seen in figure 1 and the cost function parameters for the last 300 simulations can be seen in figure 2.

$$Q = \frac{1}{15}T_s + \frac{1}{120}E \quad (18)$$

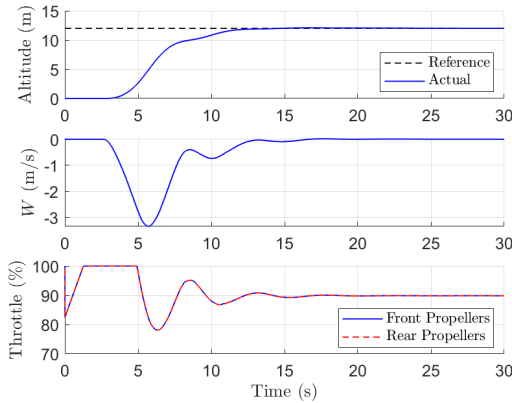


Figure 1: Controller performance for the optimised control gains. W refers to the climb rate here due to the negligible pitch or roll angles.

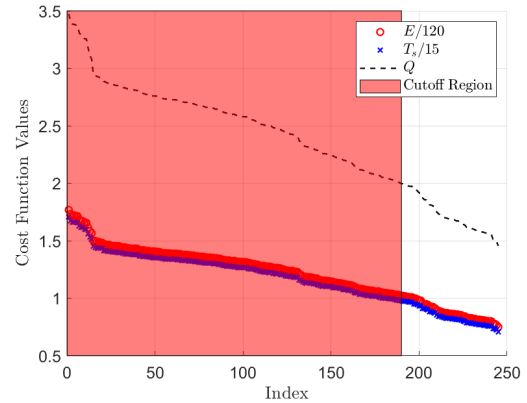


Figure 2: Sorted cost function from high to low for the last 300 simulations for the altitude controller. The optimal point is the most right point on the plot.

Table 3: Optimised control gains for the altitude controller relative to the cost function given by equation (18).

K_{p_h}	K_{i_h}	K_{d_h}	N	E (kJ)	T_s (s)	Q	Aggression (%)
0.0687	0.0116	0.1217	600	89.8	10.6	1.4558	1.59

6.3 Pitch control gains optimisation

Oscillations in pitch create two issues. Firstly, pitching reduces the amount of thrust acting vertically, causing a drop in altitude. This will cause the propellers to have to spend more energy in order to re-ascend to the target altitude. Secondly, a pitch will result in a longitudinal velocity and displacement. If the controller simply returns to zero pitch as quickly as possible, the drone will maintain this increased longitudinal velocity, slowly decelerating due to the drag of the aircraft,

where this drag is essentially equal to C_{D_0} of the aircraft. This creates an interesting optimisation: the controller must ensure that pitch angles perturbations are damped out sufficiently quickly whilst also attempting to correct for the longitudinal velocity developed, most likely by way of an overshoot. As such, the cost function in equation (19), $Q_{\Theta,1}$, is proposed, where U_s is the longitudinal velocity once a steady state pitch of 1° is reached and T_s is the time at which steady state is reached. This cost function is based on an ideal pitch correction time of 1 seconds and a steady state velocity perturbation of 3 m/s as a result of the pitch perturbation. An energy cost was chosen to not be used here as the main energy requirement is that required to maintain hover, and so minimising the time taken to correct pitch should suffice to minimise the energy used during that time period.

$$Q_{\Theta,1} = \frac{1}{1}T_s + \frac{1}{3}|U_s| \quad (19)$$

To test the pitch response, two factors are used. Firstly, the actual MTOW CG location is used, resulting in an initial pitch perturbation when full throttle is applied by the altitude controller. Secondly, a pitch rate perturbation of 0.2618 rad/s (15° 1/s) is applied for 1 second at 20 seconds. This tests the initial takeoff performance of the controller and tests its response to a pitch perturbation in hover. The results of optimising a controller utilising this cost function are shown in figures 3 and 4, with the optimum gains and resulting cost parameters being visible in table 4. The cutoff region was chosen to be a total cost of 2 as the controller must satisfy both the time and velocity requirements in the cost function, meaning that the total value for each individual cost should be less than 1. Whilst these results do show minimal perturbations in velocity and satisfactory performance in pitch, there are small oscillations in angle over a short period. These oscillations occur as the controller attempts to dampen out oscillations in velocity by way of aggressively oscillating the pitch angle, reducing the average pitch angle that the aircraft is at. Unfortunately, such a controller prescribes very aggressive throttle inputs and the oscillations in pitch angle, though small, would bring the rigid body assumption into question. To overcome this, a second cost function is proposed in equation (20), where Θ_s refers to the range in pitch values observed when steady state is reached. This cost function aims to, as before, minimise the time taken to correct pitch oscillations, but also minimise the oscillations that occur after once steady state is reached. The acceptable range for the values of steady state pitch oscillation were set to be 0.5° (0.0087 rad), which the cost function was based off. The results for this new cost function can be seen in figures 5 and 6 and table 4. As expected, this less aggressive controller does decrease the frequency of oscillations in pitch angle, but results in an increase in steady state velocity and settling of almost 2 m/s and over 1.5 seconds. Despite this reduction in recovery performance, the less aggressive control gains were chosen in order for two reasons. Firstly, a less aggressive controller is expected to better track a ramp reference signal which will be required for the transition stages. Secondly, the lower frequency of steady state pitch oscillations should make the aircraft less likely to become unstable in the event of consecutive pitch oscillations.

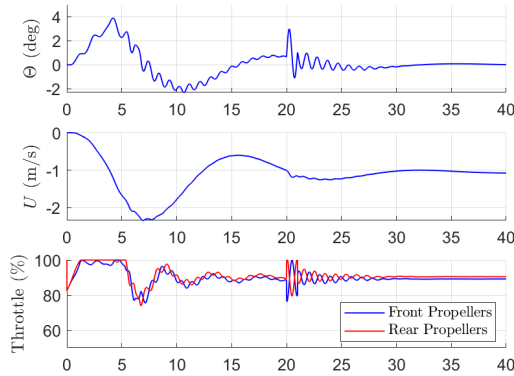


Figure 3: Pitch controller performance

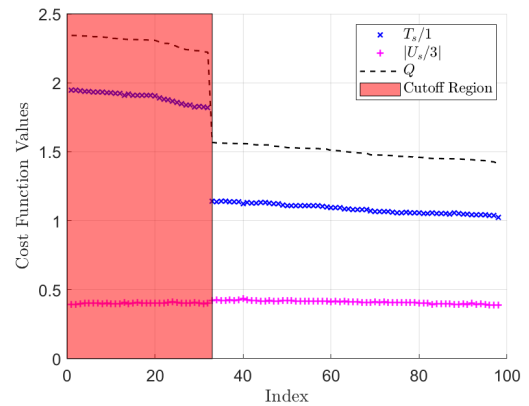


Figure 4: Sorted cost function utilising equation (19)

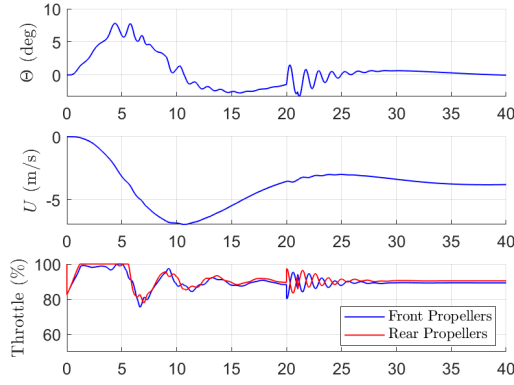


Figure 5: Pitch controller performance

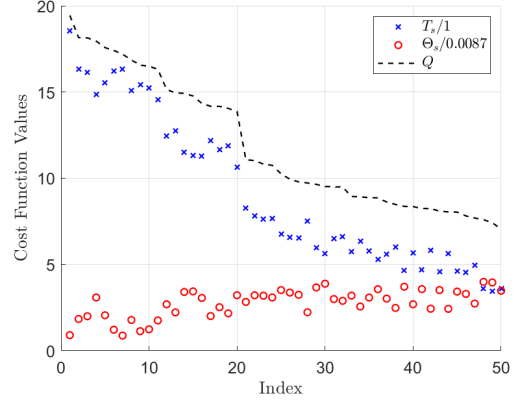


Figure 6: Sorted cost function utilising equation (20)

$$Q_{\Theta,2} = \frac{1}{1}T_s + \frac{1}{0.0087}\Theta_s \quad (20)$$

Table 4: Optimised control gains for the pitch controller relative to the cost functions $Q_{\Theta,1}$ and $Q_{\Theta,2}$ in equations (19) and (20) respectively.

Cost function	$K_{p\Theta}$	$K_{i\Theta}$	$K_{d\Theta}$	N_{Θ}	T_s (s)	U_s (m/s)	Q	Aggression (%)
$Q_{\Theta,1}$	0.1118	0.0709	0.4769	140	1.03	-1.16	1.4132	8.23
$Q_{\Theta,2}$	0.0421	0.0104	0.2396	500	3.58	-3.03	7.0512	3.83

6.4 Roll and yaw control gains

As the procedure for optimising control gains to achieve a certain performance has already been demonstrated above, a purely qualitative analysis of the lateral control gains will be conducted. There are two reasons for this. Firstly, as the lateral CG is located on the plane of symmetry of the aircraft, there are no naturally occurring lateral perturbations in the model. As such, unlike with the longitudinal CG, there are no control inputs required to sustain a zero pitch and yaw angle during the takeoff or landing. As it is assumed that the control surfaces will start to act during the transition phase as soon as possible, the rolling moment generated by the pusher propeller also does not need to be counteracted by the VTOL propellers, though this could theoretically be done if required. Secondly, unlike with the pitch controller case, any roll perturbations that are modelled would be completely arbitrary, essentially yielding arbitrary control gains. Therefore, the response of the controller to perturbations was judged from a purely visual perspective. The decision was also made to not produce yaw control gains as correcting the yaw angle is something that is not anticipated to be required as the aim is to enter forward flight as soon as possible. However, if this was ever required, it is possible to control the yaw angle as previously shown.

The roll gains were tested by applying the same 0.2618 rad/s (15 °/s) at the 20 second time point and the controller response was recorded. The gains chosen are those seen in table 5 and the performance of the controller to the applied roll perturbation can be seen in figure 7. Purely for sake of interest, the same magnitude pitch and roll oscillations used to tune the gains were applied at the same time to observe the performance of the aircraft in the event of such a perturbation. The results of this perturbation can be seen in figure 8. As can be seen, this perturbation results in an initial drop in altitude of almost 1.5 m. This is expected due to the combination of roll and pitch angle reducing the amount of thrust acting upwards. However, the response of the controller is sufficiently good to correct the pitch and roll angles quickly enough to prevent a large drop in altitude.

Table 5: Roll control gains and associated performance values as seen in figure 7.

$K_{p\Phi}$	$K_{i\Phi}$	$K_{d\Phi}$	N_Φ	T_s (s)	Aggression (%)
0.0034	0.0008	0.0095	500	14.12	0.82

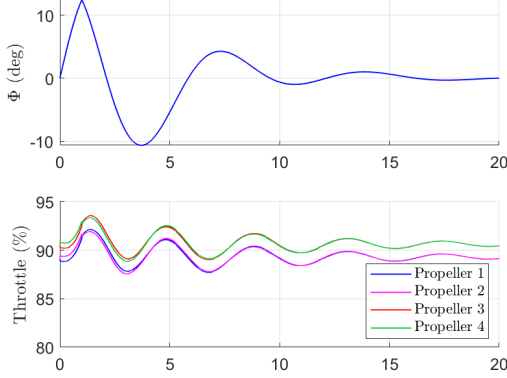


Figure 7: Roll controller performance.

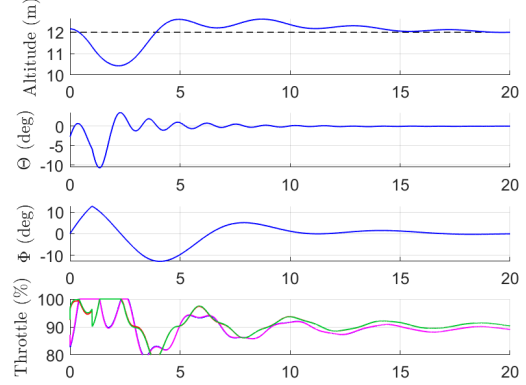


Figure 8: Response of the aircraft to a simultaneous roll and pitch perturbation. The colours for the throttle lines use the same layout as the legend dictates in figure 7.

6.5 Summary of control gains

Presented below in table 6 is a summary of all the control gains obtained above. With the controller design finalised, the flight phases can now have their viability assessed and an approximate idea of how the transition will occur can be obtained.

Table 6: Optimised control gains for the pitch controller relative to the cost functions $Q_{\Theta,1}$ and $Q_{\Theta,2}$ in equations (19) and (20) respectively.

Controller	K_p	K_i	K_d	N
Altitude	0.0687	0.0116	0.1217	600
Pitch	0.0421	0.0104	0.2396	500
Roll	0.0034	0.0008	0.0095	500

7 Results

7.1 Transition

Having optimised the all the control gains for the VTOL propellers, the transition behaviour can be evaluated. However, from transitioning from hover to forward flight, the pitch, roll and yaw authority must be handed from the VTOL propellers to the aerodynamic control surfaces. This was chosen to happen at 35 m/s. The reason for this is that the aircraft stall speed is at 29.7 m/s, but handing over control authority right at the point of stall is questionable especially if the autopilot controller ever malfunctions. As previously mentioned, the pitch angle was also set to a step reference of 0.1 degrees. This was chosen based on performance calculations which suggested that an angle of attack of 0.1 degrees is required to successfully transition [1]. The results of the transition can be seen in figure 9. As can be seen, as the aircraft accelerates, the VTOL propellers begin to approach 0% throttle as they are no longer required to maintain lift. Similarly, the controller aims to approach the pitch angle of 0.1 degrees after takeoff as can be seen by the 20 second mark. Whilst it is harder for the aircraft to achieve this pitch angle due to the natural

aerodynamic pitching behaviour, it performs sufficiently well in order to achieve a transition. As can be seen, the elevator also begins to take over once the aircraft reaches a velocity of 35 m/s which is the reason for the change in pitch behaviour.

The transition from forward flight to hover was also studied. As previously stated, the pitch controller does not directly control for velocity perturbations and therefore a reference pitch angle is supplied. This analysis seeks only to function as a proof of concept for this transition rather than a detailed control model and would need to be improved in further design iterations. Nevertheless, the results for this can be seen in figure 10. As can be seen, the aircraft pitches up and then back down in order to slow down to zero velocity. The aircraft does not quite reach zero velocity due to the nature of the controller as previously discussed. However, the fact that this manoeuvre is possible presents the case for moving forward with this design from a realistic point of view.

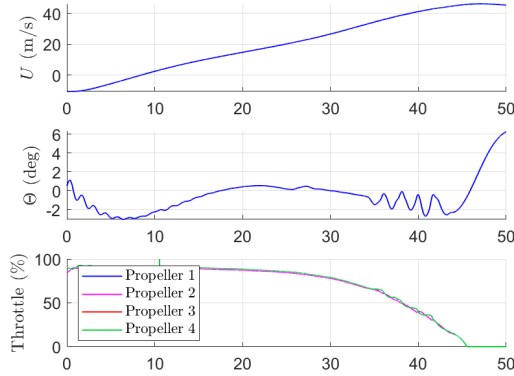


Figure 9: Performance of the aircraft in the forward transition (from hover to forward flight).

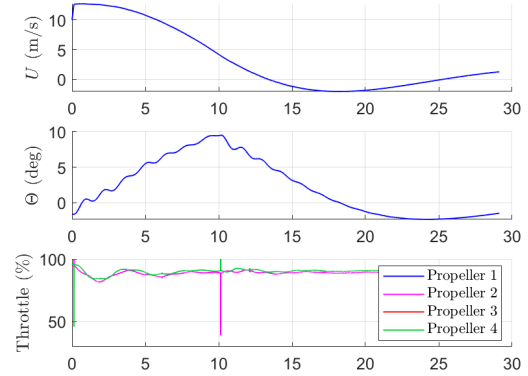


Figure 10: Performance of the aircraft in the reverse transition (from forward flight to hover).

7.2 Takeoff and landing

The takeoff phase was already analysed in detail in the optimisation of the altitude control gains, but the landing was not yet studied. As such, both the takeoff and landing flight phases are studied together here. As previously discussed, the landing aims to descend to ground at a rate of 2 m/s [1]. Utilising this as the reference signal and beginning the landing procedure at the 30 second mark, the results in figure 11 were obtained. Due to limitations in the model, the aircraft seems to continue descending at 2 m/s once it comes into contact with the ground. Whilst this is clearly not true, it can be seen that the climb rate, which corresponds to the negative of W velocity, does settle at 2 m/s as expected. The tracking of the ramp is also very non-oscillatory, suggesting that the cost functions utilised to obtain the altitude controller has successfully minimised the oscillations of the aircraft. The throttle also remains relatively constant during the landing. This is a result of a ramp signal requiring a thrust to weight ratio of 1. The constant throttle command implies that this was indeed what the controller was doing.

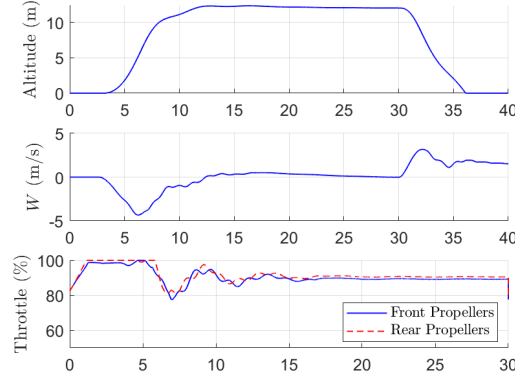


Figure 11: Performance of the aircraft in takeoff and landing.

As with the transition, these results aim to show only that the takeoff and landing is possible for this aircraft. In reality, the aircraft may need to land on a moving platform, requiring a constant pitch angle to maintain a forwards velocity as it slowly descends. The combination of the transition, takeoff and landing results however do imply that such a flight path would be achievable given certain modifications to the controller, but only under certain circumstances.

8 Conclusion

In conclusion, the VTOL flight segments of this aircraft are achievable under the current design circumstances. However, important clarifications must be made about the performance of the aircraft given the current design parameters and changes to these moving forward, specifically with regard to the robustness of the controller. If the following considerations are taken into account and either remain the same or are improved, then the results in this report serve as justification for moving forward with this design.

8.1 CG location

The longitudinal CG currently was found to be just on the verge of controllability. That is to say moving the CG even half a centimeter further back yields an aircraft which is borderline impossible to control. As the CG location tested in this report is the most extreme case relative to the centre of the propellers, it is reasonable to assume that, with current design parameters, the aircraft will perform sufficiently well. However, any change which results in the CG moving further away from the centre of the propellers will likely make the aircraft impossible to control whilst also achieving a reasonable small degree of pitch oscillations.

8.2 Motor spool-up time

The current spool up time of 5 seconds is relatively difficult to control and severely limits maneuverability. This lack of maneuverability was not found to be essential, mainly as the VTOL propellers are only being used to takeoff and the transition begins very quickly from leaving the ground. However, in the case where a hover segment is required, this problem will become a major downfall of the design especially if the aim is to quickly and accurately reach moving targets in a windy environment. It was therefore highly recommended that, in future design stages, the weight of the aircraft is reduced such that either the propeller blades can be made lighter or the battery can be made heavier in order to use a more powerful motor, especially if robust hover performance becomes a significant requirement.

References

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- [3] Elias Chimonides. Flr 05 open loop dynamic stability analysis, department of aeronautics, imperial college london, 2022.